

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-2	Team No-1	Project Title-ONLINE LEARNING MANAGEMENT SYSTEM
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1. Abstract

Traditional and semi-digital methods of managing education — scattered course materials, manual attendance, paper-based grading, and disconnected communication channels — make it difficult for institutions to deliver consistent, trackable, and scalable learning experiences to students and faculty alike.

To address this, the proposed system is built as a full-stack web application using React.js for the interface, Node.js with Express for the backend, and MongoDB/MySQL for data storage, deployed on cloud infrastructure such as AWS with Docker for containerization. It provides role-based dashboards for admins, instructors, and students, along with course management, online assessments, live/recorded classes, discussion forums, and analytics.

Together, these components digitize the end-to-end teaching-learning workflow, cutting administrative effort, improving transparency through real-time data, and giving institutions a single reliable platform that can scale from a single classroom to an entire campus.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030013	JAGADABI NANDITHA	
2520030280	TEHSEEN ASHRAFF	

3. Frameworks / Platforms / Software Used

Frontend

- React.js
- HTML5 / CSS3
- Tailwind CSS

Backend

- Node.js + Express
- REST APIs
- JWT Authentication

Database

- Cloud Storage (media)
- Redis (caching)
- MongoDB / MySQL

Cloud & DevOps

- AWS / Azure Hosting
- Docker
- GitHub CI/CD

Guide Name and Signature (remarks if any) _____